Stadistical Physics

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Índice general

Capítulo 1

Classical atomic gas

1.1. Ideal systems

In an ideal system I can express the system's hamiltonian as

$$H = \sum_i h_i \tag{1.1}$$

Applying this in the canonical ensemble framework's probability we have

$$p(C) = \frac{e^{\beta H(C)}}{Z} = \frac{e^{\beta \sum_i h_i}}{Z} = \prod_i \frac{e^{\beta h_i}}{Z} \tag{1.2}$$

Nota

The previous result is consistence with the stadistical result: if $A \cap B = 0$ then $P(A \cup B) = P(A)P(B)$

1.1.1. Examples

If we have a potential $V(r_{ij})$, this is a potential that depends on the relative position of the particle i with particles j. I cannot separate the hamiltonian because the particle i dependes on surrounding particles.

$$V(r_{ij}) \implies H \neq \sum_i h_i \tag{1.3}$$

If we have a potential $V(r_i)$, this is a potential that only depends on the particle's position. Then I can separate the hamiltonian.

$$V(r_i) \implies H = \sum_i h_i = \sum_i \frac{p_i^2}{2m_i} + \sum_i V(r_i) \tag{1.4}$$

The Ising Model that we have seen in Computational Physics has the following hamiltonian

$$H = -\frac{J}{2}\sum_{ij}^{N} s_{ij}(s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1}) \tag{1.5}$$

As we can see in the equation, the hamiltonian of the particle i-j depend on the surrounding, then we cannot express the hamiltonian as sums of each particles' hamiltonians.

$$H = -\frac{J}{2}\sum_{ij}^{N} s_{ij}(s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1}) \neq \sum_{i} h_i \tag{1.6}$$

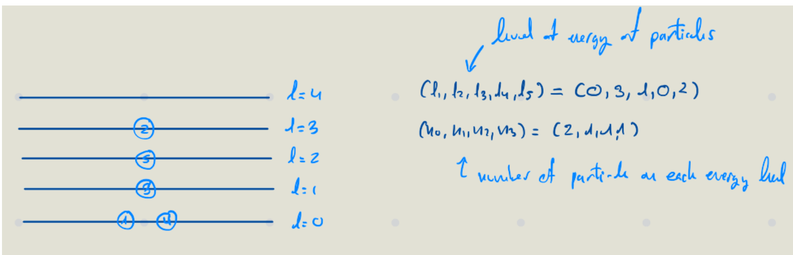
If we have the following hamiltonian, we can separate it in sum of particles' hamiltonian with $h_i = -Js_i$.

$$H = -J\sum_i s_i = \sum_i h_i = \sum_i -Js_i \tag{1.7}$$

1.2. Boltzmann Gas

The Boltzmann Gas it is the common limit of fermions and bosons.

1.2.1. In canonical ensemble



$$Z = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta E} = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta \sum_i n_i \epsilon_i} \tag{1.8}$$

Due to the condition $\sum_i n_i = N$, we cannot calculate the partition function as the sum of partition functions. But we can express the total energy as sum of energy associated to level l.

$$E = \sum_i n_i \epsilon_i = \sum_i E_{l_i} \tag{1.9}$$

Applying this in the partition function we obtain

$$Z = \sum_{l_1,\dots,l_N} e^{\beta \sum_i \epsilon_{l_i}} \tag{1.10}$$

We need take account the degeneracy. As the particle are **indistiguishable**, we can consider all the particle as the same, then we only nned to know how to distribuite N same particles in l levels.

A set of $\{n_l\}$ correspond $\frac{N!}{\prod_i n_l!}$ possible values of $\{l_1, \dots, l_N\}$.

$$Z = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta E} = \sum_{\{n_i\}, \sum_i n_i = N} \frac{\prod_i n_l!}{N!} e^{-\beta \sum_i n_i \epsilon_i} \tag{1.11}$$

1.2.2. Boltzmann approximation

The Boltzmann approximation is $n_l! \approx 1$. This is when $\rho \Lambda_T^3 \ll 1$ with $\Lambda_T^3 = h/\sqrt{2\pi m K T}$. This approximation is good when $T \upuparrows$ or/and $\rho \downdownarrows$ (very diluete cases). Under this condition, bosons and fermions **behave the same way**.

With the Boltzmann approximation, the partition function is

$$Z = \sum_i \frac{1}{N!} e^{-\beta \sum_i \epsilon_{l_i}} = \frac{Z_1^N}{N!} \tag{1.12}$$

1.2.3. Particles in a 3D potential well

Let's calculate the partition function in this system. Let's assume that

$$Z = \sum \frac{1}{N!} (Z_1)^N \tag{1.13}$$

Then let's calculate Z_1

A single particle's energy has two contribution, one due to the traslational degrees of freedom and one due to the internal degrees of freedom.

$$\epsilon = \epsilon_{tr} + \epsilon_{in} \tag{1.14}$$

Then the particle function of a single particle is

$$Z_1 = \sum_{l_{tr}} \sum_{l_{in}} e^{-\beta \epsilon_{tr} - \beta \epsilon_{in}} = Z_1^{tr} Z_1^{in} \tag{1.15}$$

In RAW we developed the solution of Schrödinger's Equation in a potential well in 1D. From this solution we obtained the expression for energy, which is related to the momentum.

$$\epsilon = \frac{nh}{4Lm} \tag{1.16}$$

Using this for the traslational energy expression

$$\epsilon_{tr} = \frac{p_x^2 + p_y^2 + p_z^2}{2m} = \frac{\Lambda}{2m} \left(\frac{n_x^2 h^2}{4L^2} + \frac{n_y^2 h^2}{4L^2} + \frac{n_z^2 h^2}{4L^2} \right) \tag{1.17}$$

Let's calculate the partition function

$$Z_1^{tr} = \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)} \tag{1.18}$$

Let's take a look how to solve individualy

$$\sum_{n=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2} n^2} \tag{1.19}$$

If we notice, p is related to the length of the well: $p = hn/2L$, then $\Delta p = p_{n+1} - p_n = h/2L$. If we consider a very large L, this is $L \rightarrow \infty$ then $\Delta p = dp$, let's use this to simplify the summatory. Let's change of variable: $n = pL/h\pi$

$$\sum_{n=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2} n^2} = \sum_{p=\frac{h}{2L}}^{\infty} e^{-\beta \frac{p^2}{2m}} \tag{1.20}$$

We introduce Δp in the summatory and we are going to perform a limit, then we can covert the summatory into an integral.

$$\lim_{\rightarrow \infty} \sum_{p=\frac{h}{2L}}^{\infty} \frac{2L}{h} \Delta p \; e^{-\beta \frac{p^2}{2m}} = \int_0^{\infty} e^{-\beta \frac{p^2}{2m}} \; dp = \frac{1}{2} \int_{-\infty}^{\infty} e^{-\beta \frac{p^2}{2m}} \; dp \tag{1.21}$$

Substituing the partition function with this integral we have

$$\frac{1}{h^3}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dp_x\, dp_y\, dp_z\, e^{-\frac{\beta}{2m}(p_x^2+p_y^2+p_z^2)} \tag{1.22}$$

Taking into account the Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}}\, dx = \sqrt{2\pi\sigma^2} \tag{1.23}$$

It follows that:

$$Z_1^{(tr)} = \frac{L^3}{h^3}(2\pi mkT)^{3/2} = \frac{V}{\Lambda_T^3} \tag{1.24}$$

Then the partition funcion is

$$Z = \frac{1}{N!}\left(\frac{V}{\Lambda_T^3}\right)^N \tag{1.25}$$

The previous result coincide with the classical ideal monoatomic gas.

1.2.4. Thermodynamic properties of the BG

$$A(T,V,N) = -kT\ln Z_{BG} = -NkT\ln z_1^{(tr)} + \underbrace{kT\ln N!}_{NkT\ln N - NkT} - NkT\ln z_1^{(in)}$$

Using the *Stirling approximation* and knowing that $z_1^{(tr)} = \frac{V}{\Lambda_T^3}$:

$$A(T,V,N) = -NkT + NkT\ln\left(\tfrac{V}{N}\right) - NkT\ln\Lambda_T^3 + NkT\ln z_1^{(in)}$$

Defining the free energy per particle $a(T,n) = \frac{A(T,V,N)}{N} = -kT + kT\ln(n\Lambda_T^3) + a_{in}(T)$, where $a_{in}(T) = -kT\ln z_1^{(in)}$ and $n = \frac{N}{V}$.

Considering $dA = -SdT - PdV + \mu dN$:

Eq. of State (Pressure)

$$P = -\left(\frac{\partial A}{\partial V}\right)_{T,N} \tag{1.26}$$

Since $-kT\ln z_1^{(in)}$ has no V dependence, only the translational degrees of freedom contribute to the equation of state (pressure):

$$P = -N\left(-\frac{\partial}{\partial V}(kT\ln(\frac{V}{N}))\right) = \frac{NkT}{V} \implies \text{Boltzmann gas behaves as an ideal gas} \tag{1.27}$$

The internal partition function $z^{(in)}$ contributes to the entropy, free energy, and specific heat.

Entropy

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,N} \implies s = -\left(\frac{\partial a}{\partial T}\right)_n = -\left[k\ln(n\Lambda_T^3) - k + a'_{in}(T) + \frac{kT\cdot 3\Lambda_T^2\cdot\Lambda'_T(T)}{\Lambda_T^3}\right] \tag{1.28}$$

Given that $\Lambda_T = \frac{h}{\sqrt{2\pi mkT}} \implies \Lambda'_T(T) = -\frac{\Lambda_T(T)}{2T}$, we derive the **Sackur-Tetrode equation**:

$$s(T,n) = k\left[\frac{5}{2} - \ln(n\Lambda_T^3)\right] - a'_{in}(T) \tag{1.29}$$

It appears that $s(T\to 0, n) \neq 0$. This occurs because we are using the BG model; therefore, T must be high and/or n must be low for the approximation to hold.

Internal Energy

$$A = U - TS \implies U = A + TS \implies u = a + Ts \tag{1.30}$$

$$u = a + Ts = \frac{3kT}{2} + a_{in}(T) - Ta'_{in}(T) \tag{1.31}$$

The $\frac{3kT}{2}$ term is the contribution from the equipartition theorem. We have a contribution of $\frac{1}{2}kT$ for each quadratic term in the Hamiltonian $H = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} + \ldots$

Specific Heat

$$c_v = \left(\frac{\partial u}{\partial T}\right)_n = \frac{3}{2}k - Ta''_{in}(T) \tag{1.32}$$

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1.2.5. Role played by internal degrees of freedom

$A_{in} = -NkT\ln z_1^{(in)}$ (Free energy)

$S_{in} = -\left(\frac{\partial A_{in}}{\partial T}\right)_{V,N} = Nk\left(\ln z_1^{(in)} + T\frac{\partial}{\partial T}\ln z_1^{(in)}\right)$ (Entropy)

$U_{in} = A_{in}+TS_{in} = -NkT\ln z_1^{(in)}+NkT\left(\ln z_1^{(in)} + T\frac{\partial}{\partial T}\ln z_1^{(in)}\right) = NkT^2\frac{\partial}{\partial T}\ln z_1^{(in)}$ (Internal ene

$C_v^{in} = \frac{\partial}{\partial T}\left[NkT^2\frac{\partial}{\partial T}\ln z_1^{(in)}\right]_{V,N}$ (Specific heat)

If $z_1^{(in)} = z_{rot}\cdot z_{vib}\cdot z_{elec}$:

$$\begin{aligned}
A^{(in)} &= -kT\ln z^{(in)} = A_{rot} + A_{vib} + A_{elec}\\
S^{(in)} &= -\left(\frac{\partial A^{(in)}}{\partial T}\right) = S_{rot} + S_{vib} + S_{elec}\\
U^{(in)} &= U_{rot} + U_{vib} + U_{elec}\\
C_v^{(in)} &= C_v^{rot} + C_v^{vib} + C_v^{elec}
\end{aligned}$$

Activation of internal energy d.o.f.

$z_1^{(in)} = \sum_n e^{-\beta\varepsilon_n^{(in)}}$ where n is the quantum number.

Spectrum of the internal d.o.f. for the internal d.o.f.

- $\Delta\varepsilon > \varepsilon_1 - \varepsilon_0$ (gap or activation energy)
- $T_a = \frac{\Delta\varepsilon}{k}$ (activation temperature)

First limit case: Thermal energy (kT) much smaller than $\Delta\varepsilon$

$kT \ll \Delta\varepsilon \implies \beta\Delta\varepsilon \gg 1$. Taking $\varepsilon_0, \varepsilon_1, \varepsilon_2\dots$ and assuming $\ln e^{-\beta\varepsilon_0} = -\beta\varepsilon_0 = -\frac{\varepsilon_0}{kT}$:

$$z^{in} = \sum_\varepsilon g(\varepsilon)e^{-\beta\varepsilon} \approx g(\varepsilon_0)e^{-\beta\varepsilon_0}\left(1 + \frac{g(\varepsilon_1)}{g(\varepsilon_0)}e^{-\beta\Delta\varepsilon} + \dots\right) \approx e^{-\beta\varepsilon_0}g(\varepsilon_0) \tag{1.33}$$

Then $C_v^{in} = k\frac{\partial}{\partial T}\left[T^2\frac{\partial}{\partial T}\ln z^{in}\right] \approx k\frac{\partial}{\partial T}\left[T^2\frac{\partial}{\partial T}\left(-\frac{\varepsilon_0}{kT}\right)\right] = k\frac{\partial}{\partial T}\left[T^2\frac{\varepsilon_0}{kT^2}\right] = 0. \implies$ In this case, internal degrees of freedom do not contribute to C_v.

Second limit case: Thermal energy (kT) much larger than $\Delta\varepsilon$

$kT \gg \Delta\varepsilon$: We can take the continuous limit of the specific energies and the system behaves as classical. $\implies$ Equipartition theorem holds $\implies U = \alpha NkT$ ($C_v = \text{constant}$).

Graph of $C_v = C_v^{(tr)} + C_v^{(in)}$

- At low T: $C_v^{in} = 0$, so $C_v = \frac{3}{2}k$.

- At high T ($kT \gg \Delta\varepsilon$): C_v increases and reaches a new plateau due to the activation of internal degrees of freedom.

To study the vibrational d.o.f., we are going to study diatomic molecules.

$V(r)$ Taylor expansion around r_0, the equilibrium position:

$$V(r) \approx V(r_0) + \cancel{V'(r_0)(r-r_0)} + \frac{1}{2}V''(r_0)(r-r_0)^2 + \cdots \implies \frac{1}{2}kx^2 \rightarrow \text{Harmonic Oscillator} \tag{1.34}$$

Now we consider Schrödinger eq:

$$\hat{H}\psi = E\psi \rightarrow -\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2} + \frac{1}{2}kx^2\psi = E\psi \rightarrow \omega = \sqrt{\frac{k}{\mu}} \rightarrow \frac{d^2\psi}{dx^2} + \frac{2\mu}{\hbar^2}E = \dots \tag{1.35}$$

The solution to this is: $\varepsilon_n = \hbar\omega(n + 1/2)$ with $\omega = \sqrt{\frac{k}{\mu}}$, where $n = 0, 1, 2, 3\dots$

Partition function:

$$z_1^{vib} = \sum_{n=0}^{\infty} e^{-\beta\varepsilon_n} = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+1/2)} = e^{-\frac{\beta\hbar\omega}{2}} \sum_{n=0}^{\infty} (e^{-\beta\hbar\omega})^n = e^{-\frac{\beta\hbar\omega}{2}} \frac{1}{1-e^{-\beta\hbar\omega}} = \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1-e^{-\beta\hbar\omega}} \cdot \frac{e^{\frac{\beta\hbar\omega}{2}}}{e^{\frac{\beta\hbar\omega}{2}}} = \frac{\quad}{e^{\frac{\beta\hbar\omega}{2}} -} \tag{1.36}$$

Using the identity $2\sinh(x) = e^x - e^{-x}$:

$$z_1^{vib} = [2\sinh(\beta\hbar\omega/2)]^{-1} \implies Z^{vib} = (z_1^{vib})^N = [2\sinh(\beta\hbar\omega/2)]^{-N} \tag{1.37}$$

Free energy: $A_{vib} = -kT\ln Z = N[\frac{1}{2}\hbar\omega + kT\ln(1-e^{-\beta\hbar\omega})] \implies$ does not depend on V.

Therm. pressure: $P = -(\frac{\partial A}{\partial V})_{T,N} = 0 \implies$ vibrational d.o.f. do not contribute to the pressure.

Entropy: $s_{vib} = -(\frac{\partial a}{\partial T})_n = Nk\left[\frac{\beta\hbar\omega}{e^{\beta\hbar\omega}-1} - \ln(1-e^{-\beta\hbar\omega})\right]$

Internal energy: $u_{vib} = a + Ts = Nk\left[\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\beta\hbar\omega}-1}\right]$

Specific heat: $c_{v,vib} = \frac{1}{N}(\frac{\partial U}{\partial T})_{V,N} = k(\beta\hbar\omega)^2\frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}$

As we see, the z^{vib} which is an internal degree of freedom does not contribute to the state of stage equation (pressure).

$\Theta_v \rightarrow$ *vibrational activation temperature* Activation energy: $\varepsilon_n = \hbar\omega(n + 1/2)$ $k\Theta_v = \hbar\omega \implies \Theta_v = \frac{\hbar\omega}{k} \implies c_{v,vib} = k(\frac{\Theta_v}{T})^2\frac{e^{\Theta_v/T}}{(e^{\Theta_v/T}-1)^2}$

We shall distinguish two cases:

- $T \gg \Theta_v \implies \frac{\Theta_v}{T} \ll 1$: $c_{v,vib} \approx k(\frac{\Theta_v}{T})^2 \frac{1+\dots}{(1+\frac{\Theta_v}{T}+\dots-1)^2} = k\frac{(\Theta_v/T)^2}{(\Theta_v/T)^2} = k$
- $T \ll \Theta_v \implies \frac{\Theta_v}{T} \gg 1$: $c_{v,vib} \approx k(\frac{\Theta_v}{T})^2 \frac{e^{\Theta_v/T}}{(e^{\Theta_v/T})^2} \xrightarrow{T\to 0} 0$

Rotational d.o.f.

Rigid rotor model: $T = \frac{1}{2}I\omega^2$ (rotational kinetic energy) with $I = \mu r_0^2$ and $\mu = \frac{m_1 m_2}{m_1+m_2}$. $L = I\omega \implies \vec{L} = \vec{r} \times \vec{p}$, then $T = \frac{L^2}{2I}$.

$\hat{H}\psi = E\psi \to \frac{L^2}{2I}\psi = E\psi$ $\vec{L} = \vec{r} \times \vec{p} = \cdots \implies L^2 = L_x^2 + L_y^2 + L_z^2$

Using spherical coordinates: $x = r\sin\theta\cos\phi$ $y = r\sin\theta\sin\phi$ $z = r\cos\theta$

$L^2 = -\hbar^2 \left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$

The solution of this Schrödinger equation is: $\psi(\theta,\phi) = AY_{lm}(\theta,\phi) = Y_{lm}(\theta,\phi)$ where $l = 0, 1, 2\dots$ and $m = -l, \dots, l$.